The startup database vendor relies on repeat usage for a very different reason. Database technology is used only as the foundation for a customer's own products, such as a website or a point of sale system. Once you build a product on top of a particular database technology, it is extremely difficult to switch. In the IT industry, such customers are said to be locked in to the vendor they choose. For such a product to grow, it has to offer such a compelling new capability that customers are willing to risk being tied to a proprietary vendor for a potentially long time.

Thus, both businesses rely on having a high customer retention rate. They have an expectation that once you start using their product, you will continue to do so. This is the same dynamic as a mobile telephone service provider: when a customer cancels his or her service, it generally means that he or she is extremely dissatisfied or is switching to a competitor's product. This is in contrast to, say, groceries on a store aisle. In the grocery retail business, customer tastes fluctuate, and if a customer buys a Pepsi this week instead of Coke, it's not necessarily a big deal.

Therefore, companies using the sticky engine of growth track their attrition rate or churn rate very carefully. The churn rate is defined as the fraction of customers in any period who fail to remain engaged with the company's product.

The rules that govern the sticky engine of growth are pretty simple: if the rate of new customer acquisition exceeds the churn rate, the product will grow. The speed of growth is determined by what I call the rate of compounding, which is simply the natural growth rate minus the churn rate. Like a bank account that earns compounding interest, having a high rate of compounding will lead to extremely rapid growth—without advertising, viral growth, or publicity stunts.

Unfortunately, both of these sticky startups were tracking their progress using generic indicators such as the total number of customers. Even the actionable metrics they were using, such as the activation rate and revenue per customer, weren't very helpful because in the sticky engine of growth, these variables have little impact on growth. (In the sticky engine of growth, they are better